

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: VIII

Subject Name: Advanced Instrumentation Techniques

Subject Code: BP811ET

Max. Marks: 75

Date: 21/07/2022

Duration: 3.45 Hrs.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q. 1. Multiple Choice Questions (MCQs) = $20 \times 1 = 20$ (All the questions are compulsory)

i) NMR Spectroscopy is used for determining structure in which of the following materials?

- | | |
|--------------------------|---------------------------------|
| a) Radioactive materials | b) Insoluble chemical compounds |
| c) Liquids | d) Gases |

ii) Greater the electron negativity of the atom..... the deshielding effect

- | | |
|----------------|------------|
| a) Lesser | b) Greater |
| c) Remain same | d) None |

iii) Highest m/z peak in mass spectrum is called as -----

- | | |
|------------------|------------------|
| a) Base peak | b) Fragment peak |
| c) Isotopic peak | d) Parent peak |

iv) If the number of neutrons and the number of protons are both even, then the nucleus has

- | | |
|----------------------|-----------------|
| a) Half-integer spin | b) Integer spin |
| c) NO spin | d) None |

v) Which of the following is the type of TGA except?

- | | |
|---------------------------|--------------------|
| a) Isothermal / static TG | b) Quasistatic TGA |
| c) Dynamic TGA | d) None |

vi) The Rotating crystal method is mainly used to determine

- | | |
|---------------------|--------------------|
| a) Lattice constant | b) Lattice pattern |
|---------------------|--------------------|



c) Orientation in large single crystals

d) None

vii) X- rays are -----

a) Deflected by electric field but not by a magnetic field

b) Deflected by magnetic field but not by an electric field

c) Deflected by both a magnetic field and an electric field

d) Not deflected by an electric field or a magnetic field

viii) In technique sample is heated to a constant weight at each of a series of increasing temperature

a) Isothermal / static TG

b) Quasistatic TGA

c) Dynamic TGA

d) None

ix) In a measurement, what is the term used to specify the closeness of two or more measurements?

a) Precision

b) Accuracy

c) Robustness

d) Specificity

x) With respect to calibration of uv-vis spectrophotometer, which of the following material is used to evaluate the control of absorbance

a) Sodium thiosulphate

b) Potassium dichromate

c) Holmium oxide

d) Potassium permanganate

xi) ICH stands for -----

a) International Council for Harmonization

b) International Conference for Harmonization

c) International Council for Hominization

d) International Conference for Hominization

xii) The lowest amount of analyte in a sample which can be detected and quantified is called as?

a) Limit of Detection

b) Accuracy

c) Limit of Quantitation

d) Specificity

xiii) Which of the following is type of immunoassay except?

a) Radio immune assay

b) Enzyme Immunoassay

c) Pro Immunoassay

d) All of the above

xiv) In solid phase extraction eluting solvent added to ----- analyte selectively

a) Adsorb

b) Desorb

c) Both

d) None of the above

xv) How many different types of radioimmunoassay are -----

a) Four

b) Two

c) One

d) Three

xvi) Solvent extraction is better if repeated extraction is done using-----

a) Large solvent

b) Small Solvent

c) Extra solvent

d) Normal solvent

xvii) Which of the following is most often used in the chromatograph in gas chromatograph MS?

a) Curvette

b) Paper support

c) Capillary tube

d) Flask

xviii) Thermospray is used as an interface in-----

a) GC-MS

b) HPLC-MS

c) GC-FTIR

d) HPLC-FTIR

xix) Which of the following is the type of separator used in commercial GC-MS systems?

a) Jet type molecular separator

b) Porous tube

c) Teflon tube

d) Flow type separator

xx) GC- MS has been developed for which of the following systems?

a) Packed column

b) Open tubular column

c) Capillary column

d) Porous layer column

Q. 2. Long Answers) = 2 x 10 = 20 (Answer 2 out of 3)

i) Explain in detail chemical shift and spin-spin coupling

ii) Describe in detail Thermogravimetric analysis

iii) Define Validation and explain it in detail as per ICH guidelines

Q. 3. Short Answers = 7 x 5 = 35 (Answer 7 out of 9)

i) What are the applications of NMR?

ii) Discuss any two diffraction methods used in X-ray diffraction

iii) Explain calibration of UV-Visible spectrophotometer

iv) Write a short note on Radioimmune assay

v) Write a short note on GC-MS/MS

vi) Explain LC-MS/MS

vii) Explain continuous extraction type in liquid-liquid extraction

viii) Describe Differential Thermal Analysis

ix) What is the importance of Radioimmune assay? Enlist types & Explain applications of

Radioimmune assay (5M)

-----END OF THE PAPER-----

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Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) What is Chemical shift? Enlist the factors affecting Chemical shift.
- ii) Write down the principle of Mass spectroscopy. Enlist different ionizers used in MS.
- iii) Differentiate between TGA and DTA.
- iv) Explain Bragg's law. Enlist the different X-ray Diffraction techniques.
- v) Define Distribution Coefficient and Distribution Ratio.
- vi) What is mean by Hyphenated Technique? Classify it.
- vii) Explain applications of RIA.
- viii) Differentiate between Calibration and Validation.
- ix) Enlist the ionization techniques used in LC-MS/MS.
- x) What do you mean by Spin- spin coupling? Write applications of NMR spectroscopy.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Explain in brief, "Analytical methods of validation as per ICH Guidelines"
- ii) Explain in detail Calibration of HPLC and electronic balance.
- iii) Enlist different methods of thermal analysis. Explain Thermogravimetric analysis in detail

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Write a short note on H^1 NMR.
- ii) Explain X-ray Diffraction techniques in brief.
- iii) Describe GC-MS MS with principle and applications.
- iv) Explain the calibration of UV spectrophotometer.
- v) Write a short note on Differential Scanning Calorimetry.
- vi) Describe the procedure and applications of RIA.
- vii) Explain fragmentation patterns in Mass Spectrometry.
- viii) What is mean by extraction technique. Explain solid phase extraction.
- ix) Write a short note on MALDI.

-----END OF THE PAPER-----

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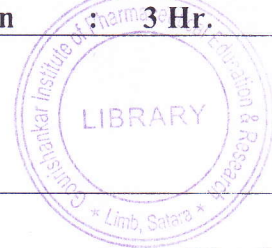
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Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Why TMS is used as reference standard in NMR spectroscopy?
- ii) Define calibration and validation as per ICH guidelines.
- iii) What are hyphenated techniques? State their importance.
- iv) Write the number of signals for following compounds?
1. $\text{CH}_3\text{-CH}_2\text{-CHBr-COOH}$ 2. $\text{Cl-CH}_2\text{-CH}_2\text{-CH}_2\text{-Br}$
- v) What is origin of X-ray? State Bragg's equation.
- vi) What is liquid-liquid extraction? Enlist types of liquid-liquid extraction methods.
- vii) Describe base peak and molecular ion peak?
- viii) What is shielding and deshielding effect?
- ix) Write principle of C-NMR.
- x) State importance of ionization in mass spectrometry. Enlist ionization techniques used in mass spectrometry.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) What is basis of thermal method of analysis? Enlist different methods of it. Explain Thermogravimetric analysis in detail.
- ii) Explain in brief principle, instrumentation and applications of mass spectrometry.
- iii) Discuss principle, different methods and applications of x-ray diffraction methods.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) What is radioimmunoassay? Give procedure and applications of RIA.
- ii) Explain in detail GC-MS/MS.
- iii) Write a short note on DTA.
- iv) Elaborate fragmentation pattern of hydrocarbons in mass spectrometry.
- v) Discuss calibration of UV-Visible spectrophotometer
- vi) Describe chemical shift. Explain factors affecting chemical shift.
- vii) Explain any two interfaces of LC-MS/MS
- viii) Write a short note on spin-spin coupling and coupling constant.
- ix) Explain principle, procedure and applications of solid-phase extraction.

-----END OF THE PAPER-----